

Davis Ernst

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EDUCATION

University of Saskatchewan

Bachelor of Science in Computer Science

Saskatoon, SK

Expected April 2027

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C, SQL, HTML/CSS

Frameworks: Next.js, React, FastAPI, Node.js, Tailwind CSS

Data & Infrastructure: PostgreSQL, Docker, WebSockets

Developer Tools: Git, Linux, Bash

PROJECTS

Real-Time Game Analysis Application | *Python, FastAPI, Next.js*

- Engineered a real-time analysis platform through the University of Saskatchewan Sports Engineering Club, actively used by the women's soccer team to give coaching staff live performance metrics and visual overlays
- Built the Next.js interface used by the team to view analytical output, backed by a FastAPI service that streams object and keypoint-detection results over WebSockets
- Partnered with coaches and club members to translate performance-analysis needs into technical specifications, gathering feedback to continuously refine detection accuracy and UI features

Game Boy Emulator | *TypeScript, Node.js, Systems Programming*

- Implemented the complete Sharp LR35902 CPU instruction set with accurate flag behavior and instruction timing, plus a scanline-based PPU renderer
- Added MBC0/1/3/5 memory banking and compatibility with 50+ commercial ROMs; validated behavior with standard test-ROM suites in a headless Node.js environment
- Published a zero-runtime-dependency npm package for browser and Node.js applications

Shadow Basketball Registration Platform | *TypeScript, Next.js, PostgreSQL*

- Built and deployed a full-stack tryout registration platform for a local basketball club, including public registration and an administrative dashboard for session and capacity management
- Implemented authentication, role-based access control, and a PostgreSQL schema with migrations for scheduling and registration data

EXPERIENCE

Technical Operations Summer Student

Innovation Saskatchewan

May 2026 – Aug. 2026

Saskatoon, SK

- Supported building operators and tradespeople with preventative maintenance, repairs, equipment installations, and changing operational priorities across park facilities
- Conducted mechanical-room walkthroughs, completed work orders, monitored safety equipment, and maintained inventories of tools, parts, and supplies

General Labourer

Wright Construction

June 2025 – Aug. 2025

Saskatoon, SK

- Supported construction crews through material handling and site logistics while maintaining a safe, organized worksite

Sales Associate

Northern Factory Workwear

June 2024 – Sept. 2024

Saskatoon, SK

- Advised tradespeople and industrial clients on safety equipment and workwear, providing accurate recommendations in a specialized retail environment